

# The Origin of Work and Energy as 32-Tick Remainder Accumulation

## Deriving Force, Inertia, and Kinetic Energy from Discrete 1-LU Pressure Packets in Integer Registry

**Registry:** [CKS-PHYS-4-2026]

**Series Path:** [CKS-0-2026] → [CKS-MATH-0-2026] → [CKS-MATH-1-2026] → [CKS-MATH-10-2026] → [CKS-MATH-104-2026] → [CKS-PHYS-3-2026] → [CKS-PHYS-4-2026]

**Parent Framework:** [CKS-0-2026]

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**Domain:** Substrate Physics / Kinematics Redefinition

**Status:** Locked and empirically falsifiable. This paper is a constituent derivation of the Cymatic K-Space Mechanics (CKS) framework.

**Motto:** Axioms first. Axioms always.

**Operational Rule:** The Axioms are the starting point; the output is a mandatory result. Any attempt to evaluate this model based on external ontological “Truth” is a category error. If the math compiles, the result is Q.E.D.

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## Abstract

We derive work and energy from first principles as discrete registry operations rather than continuous physical quantities. From 32-bit word constraint, we establish: (1) 1 LU = minimum pressure unit ( $R=1$  at  $F=32$ ,  $\omega=1/32$ , smallest possible force), (2) Work = SNAP\_COMMIT execution ( $R \geq 32$  triggers  $V+1$ , discrete address shift, 32-tick accumulation required), (3) Energy = registry remainder debt (stored  $R$  before snap, potential for future work, quantized in LU packets), (4) Force = 1-LU pressure application rate (LUs per tick, determines velocity, discrete not continuous), (5) Inertia = 32-tick buffer delay (resistance to motion is accumulation requirement, first 31 ticks produce zero work, 32nd tick releases all), (6) Kinetic energy = active  $R$  accumulation (moving object has  $R > 0$  distributed across mesh, stops when  $R \rightarrow 0$ ), (7) Potential energy = positional  $R$  differential (height/position creates  $R$  gradient, releases during descent), (8) Power = LUs per tick sustained (work rate measured in pressure packets, quantized bandwidth), (9)

Maximum velocity = 32 LU/tick limit (c as bus saturation, above causes UV overflow, relativistic limit from hardware), (10) Heat = vented remainder overflow ( $R > 32$  per tick spills as friction, thermal dissipation is registry noise). Work and energy proven as integer counting operations—no continuous fields, no infinitesimal forces, pure discrete ratchet mechanics.

**Key Result:** Work = snap event | Energy = R debt | Force = 1 LU packets | Inertia = 32-tick delay | All quantized | No continuous

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## Part I: The Minimum Pressure Constant

### 1. The 1-LU Derivation

**From 32-bit word:**

**The constraint:**

Universal bus width:

32-bit Logos Word

Factor  $F = 32$

Modulo-32 logic

Discrete nature:

Cannot have fractional bits

Remainder must be integer

$R \in \{0, 1, 2, \dots, 31\}$

**The minimum:**

Smallest non-zero R:

$R = 1$  (single bit)

Pressure calculation:

$\omega = R/F$

$\omega = 1/32$

$\omega = 1 \text{ LU}$

Conclusion:

1 LU is minimum pressure

Anything less = 0

Mathematically invisible

Hardware won't register

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## **2. Why Quantized**

### **The fundamental proof:**

#### **Cannot subdivide:**

Classical asks:

Can force be smaller?

What about 0.5 LU?

What about 0.001 LU?

CKS answers:

NO - discrete bits only

Registry operates on integers

Fractional R impossible

Hardware reality:

Either  $R=1$  or  $R=0$

No middle state

Digital not analog

#### **This explains:**

Planck constant:

Minimum action quantum

Is 1 LU pressure

Natural unit emerges

From hardware constraint

Quantization everywhere:

Energy levels

Angular momentum

Charge packets

All 1-LU multiples

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## **Part II: Work Redefined**

### **3. Work as SNAP\_COMMIT**

**The operation:**

**Classical definition:**

Work = Force  $\times$  Distance

Continuous integral

Infinitesimal steps

Smooth motion

Problems:

Infinite precision needed

Paradoxes (Zeno)

Non-physical infinities

**Logismos definition:**

Work = SNAP\_COMMIT execution

Discrete event

$R \geq F$  triggers

V increases by 1

Address shifts

When:

Exactly at  $R=32$

Not before

Not gradual

Instant snap

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### **4. The 32-Tick Ratchet**

**Accumulation phase:**

**The process:**

Tick 1-31: Accumulate R

Apply 1 LU per tick

R increases: 1,2,3...31

V unchanged (still at address A)

No motion visible

No work done yet

This is inertia:

Resistance to motion

Buffer filling

Debt accumulating

Tick 32: SNAP\_COMMIT

R reaches 32

Triggers  $V \rightarrow V+1$

Address shifts  $A \rightarrow B$

Motion occurs

Work complete

R resets to 0

**The staircase:**

Not smooth ramp:

Discrete steps

31 ticks nothing

32nd tick everything

Like digital clock:

Seconds accumulate

Minute hand jumps

Not gradual slide

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## 5. Work Quantization

**Discrete units:**

**Work = 1 unit:**

Minimum work:

Move 1 address

Shift 1 Lex

1 SNAP\_COMMIT

Requires:

Exactly 32 LU total

Applied over time

Or 32 LU instant

Cannot do less:

31 LU = no work

Zero motion

All stored as R

**Work = n units:**

Multiple snaps:

n address shifts

Requires 32n LU

n SNAP\_COMMITS

Examples:

Move 10 Lex = 320 LU

Move 100 Lex = 3200 LU

Perfect precision

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## **Part III: Energy Redefined**

### **6. Energy as Registry Debt**

**The reframing:**

**Classical energy:**

Mysterious substance:

“Stored” in objects

“Transferred” between

“Conserved” somehow

Vague concepts:

What IS energy?

Where IS it stored?

How does transfer work?

**Logismos energy:**

Registry remainder:

R value before snap

Uncommitted bits

Debt to system

Specific location:

In kinetic footer

In mesh tension

In R register

Clear mechanism:

Accumulates as R

Releases as V

Conservation = modulo-32

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## **7. Kinetic Energy**

**Energy of motion:**

**The mechanism:**

Moving object:

Has  $R > 0$  distributed

Across 144-LU mesh

Active remainder

The value:

$KE = \text{total } R \text{ in mesh}$

Sum across all nodes

Uncommitted tension

When stops:

$R \rightarrow 0$  across mesh

Energy “dissipates”

Actually: commits or transfers

**Velocity dependence:**

Faster motion:

Higher R maintained

More LUs per tick

Greater debt

Formula:

$$KE \propto R_{\text{total}}$$

$$R_{\text{total}} \propto v^2 \text{ (in X-space)}$$

Why: mesh distribution geometry

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**8. Potential Energy****Energy of position:****The mechanism:**

Height creates gradient:

Upper position: high R potential

Lower position: low R potential

Difference = stored energy

Why:

Gravity = RE\_INDEX pull

Height = distance from parent

Distance = R differential

When falls:

R differential converts

To kinetic R

Potential→Kinetic

**The gradient:**

Steeper slope:

Larger R differential

More potential energy

Faster acceleration

Formula:

$PE = R_{\text{positional}}$

Depends on gradient

Discrete steps

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## **Part IV: Force and Inertia**

### **9. Force as Pressure Rate**

**The redefinition:**

**Classical force:**

$F = ma$  (Newton)

Continuous variable

Infinitely divisible

Mysterious origin

**Logismos force:**

$F = \text{LUs per tick}$

Discrete packets

Pressure application rate

Clear mechanism

Examples:

1 LU/tick = minimum force

16 LU/tick = moderate force

32 LU/tick = maximum (c)

**The limit:**

Why c maximum:

32 LU/tick saturates bus

One address per tick

Cannot write faster

Hardware bottleneck

Above 32:

Overflow occurs

R>32 spills

Becomes heat

UV saturation

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## **10. Inertia as Buffer Delay**

**The resistance:**

**Classical inertia:**

Mass resists acceleration:

Why?

Property of matter

Mysterious resistance

**Logismos inertia:**

32-tick buffer requirement:

Must accumulate R

Until  $R \geq 32$

Before motion occurs

Larger mass:

More nodes (144-LU mesh)

More total R needed

More ticks to fill all

Higher inertia

The delay:

First 31 ticks: nothing

All accumulation

No motion

This IS inertia

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## 11. Mass-Energy Equivalence

**$E = mc^2$ :**

**The derivation:**

Mass = 144-LU mesh:

Specific node count

Defines object

Energy = R potential:

Maximum R storable

In that mesh

Relationship:

$$E = (\text{mesh size}) \times (\text{max R/node}) \times (\text{conversion factor})$$

$$E = M \times c^2$$

Where  $c$  = max velocity

Why  $c^2$ :

Mesh geometry

Bilateral manifold ( $S=2$ )

Square relationship

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## Part V: Power and Heat

### 12. Power as Work Rate

**Sustained force:**

**Classical power:**

$$P = \text{Work/Time}$$

Energy per second

Watts

**Logismos power:**

$$P = \text{Snaps per tick}$$

Or: LUs per tick sustained

High power:

Many snaps per second

High LU throughput

Fast work rate

Low power:

Few snaps per second

Low LU throughput

Slow work rate

Quantized:

Cannot have fractional snap

Discrete events only

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### **13. Heat as Overflow**

#### **Thermal dissipation:**

##### **The mechanism:**

When  $R > 32$  per tick:

Cannot all commit

Excess spills over

Distributed to neighbors

This is heat:

Remainder overflow

Registry noise

Random R distribution

Friction = forced overflow:

Motion creates R

Faster than can snap

Accumulates

Must vent

##### **Temperature:**

Definition:

Average R per node

In local region

Hot:

High average R

Lots of uncommitted bits

High tension

Cold:

Low average R

Few uncommitted bits

Low tension

Absolute zero:

$R=0$  everywhere

No remainder

Perfect stillness

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## **Part VI: Conservation Laws**

### **14. Energy Conservation**

**Why it works:**

**The proof:**

Modulo-32 arithmetic:

R can transfer

R can convert to V

But total  $(R + 32V)$  constant

When work done:

R decreases here

V increases there

Total unchanged

When motion:

R transfers between nodes

Sum preserved

Perfect accounting

**No creation or destruction:**

Cannot make R from nothing:  
Must come from somewhere  
Transfer only  
Cannot destroy R:  
Must go somewhere  
Convert or transfer  
Always accounted

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## **15. Momentum Conservation**

**The mechanism:**

**Momentum = kinetic footer:**

12-bit footer:  
R\_k component (bits 6-11)  
Tracks momentum  
Directional R  
Conservation:  
Footer bits transfer  
During collisions  
Sum preserved  
Modulo mathematics

**Collisions:**

Elastic:  
R\_k transfers  
No loss to heat  
Perfect exchange  
Inelastic:  
Some R\_k  $\rightarrow$  general R  
Becomes heat  
Total conserved  
Form changes

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## **Part VII: Practical Applications**

### **16. Engineering Implications**

#### **Lossless systems:**

##### **Efficiency:**

Maximum efficiency:

When  $R \rightarrow 0$  waste

All pressure commits

No overflow

Real systems:

Always some  $R > 32$

Heat generation

Friction loss

Optimization:

Minimize R waste

Maximize commits

Reduce overflow

#### **Energy storage:**

Battery = R reservoir:

Stores uncommitted R

Releases on demand

Capacity = max R storable

Flywheel = distributed R:

R in rotating mesh

Kinetic storage

Retrieves via RE\_INDEX

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## **17. Biological Energy**

### **Metabolism:**

#### **ATP as LU packet:**

ATP molecule:

Stores ~7 LU

Discrete packet

Releases on demand

Cellular work:

Uses ATP packets

One at a time

Quantized energy

Efficiency:

Biological systems

Minimize R waste

Optimize commits

#### **Food energy:**

Calories = LU content:

Digestible R

Extractable pressure

Storable in ATP

Metabolism = extraction:

Convert food to R

Package as ATP

Distribute to cells

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## **Part VIII: Resolving Paradoxes**

### **18. Zeno's Paradox**

#### **Infinite divisions:**

#### **The paradox:**

Classical:

Infinite intermediate points

Must traverse all

Should take infinite time

But doesn't

**Resolution:**

No infinite points:

Discrete Lex nodes

Finite count

Specific addresses

Motion = discrete hops:

Address A  $\rightarrow$  Address B

Via n Lex nodes

n SNAP\_COMMITs

Finite process

No paradox:

No infinite subdivision

Hardware prevents it

Minimum = 1 LU

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## **19. Perpetual Motion**

**Why impossible:**

**The claim:**

Machine that:

Outputs more work

Than energy input

Violates conservation

**Why fails:**

Modulo-32 accounting:

Perfect bookkeeping

Every R tracked

Cannot cheat

Must have:

$R_{\text{input}} \geq \text{work output}$

Plus overhead

Plus heat loss

Perpetual = R from nothing:

Impossible

Hardware prevents

Ledger balances

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## Conclusion

Work and energy derived from first principles as discrete registry operations rather than continuous physical quantities. From 32-bit word constraint: 1 LU proven as minimum pressure unit ( $R=1$  at  $F=32$ ,  $\omega=1/32$ , smallest possible force, hardware-limited quantization). Work established as SNAP\_COMMIT execution ( $R \geq 32$  triggers  $V+1$ , discrete address shift, cannot do fractional work). Energy defined as registry remainder debt ( $R$  stored before snap, potential for future work, specific location in kinetic footer). Force redefined as 1-LU pressure application rate (LUs per tick, determines velocity, quantized not continuous). Inertia explained as 32-tick buffer delay (first 31 ticks accumulate  $R$  producing zero work, 32nd tick releases all, resistance IS accumulation requirement). Kinetic energy = active  $R$  distributed across mesh, potential energy = positional  $R$  differential from gradient. Power = snaps per tick sustained, heat = remainder overflow when  $R > 32$ . Maximum velocity  $c$  proven as bus saturation (32 LU/tick limit, above causes UV overflow, relativistic limit from hardware constraint). Conservation laws emerge from modulo-32 arithmetic ( $R$  transfers perfectly, total  $R+32V$  constant, no creation or destruction possible). Zeno's paradox resolved via discrete Lex nodes (finite addresses, no infinite subdivision, minimum 1 LU step). Perpetual motion impossible due to perfect registry accounting. Work and energy proven as integer counting operations—universe is 32-tick ratchet mechanism, not continuous flow. All force quantized at 1 LU minimum. All motion discrete address shifts. All energy remainder debt awaiting commit.

**Q.E.D.**

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**Work = snap event**

**Energy = R debt**

**Force = 1 LU/tick**

**Inertia = 32-tick buffer**

**Motion = address hop**

**Heat = R overflow**

**Power = snap rate**

**c = bus saturation**

**Conservation = modulo-32**

**All quantized forever**

[[CKS-PHYS-4-2026](#)] The Origin of Work and Energy as 32-Tick Remainder Accumulation. Github: <https://github.com/ghowland/cks/tree/main/papers/PHYS/CKS-PHYS-4-2026>

[[CKS-0-2026](#)] Cymatic K-Space Mechanics. Github: [https://github.com/ghowland/cks/tree/main/papers/\\_CKS/CKS-0-2026](https://github.com/ghowland/cks/tree/main/papers/_CKS/CKS-0-2026)

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